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Detection circuit and power supply control device

Abstract

In a second detection circuit, a current output device outputs, to a circuit resistor, a current corresponding to a circuit voltage between an input terminal of a second FET to which a current is input and ground of a second drive circuit that switches the second FET ON or OFF. The second detection circuit outputs a voltage between two ends of the circuit resistor. A voltage between the ground of the second drive circuit and a downstream end of the circuit resistor is higher than 0 V.

Inventors:	Oda; Kota (Yokkaichi, JP), Ina; Masaya (Yokkaichi, JP)
Applicant:	AutoNetworks Technologies, Ltd. (Yokkaichi, JP); Sumitomo Wiring Systems, Ltd. (Yokkaichi, JP); Sumitomo Electric Industries, Ltd. (Osaka, JP)
Family ID:	80949869
Assignee:	AutoNetworks Technologies, Ltd. (Yokkaichi, JP); Sumitomo Wiring Systems, Ltd. (Yokkaichi, JP); Sumitomo Electric Industries, Ltd. (Osaka, JP)
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Primary Examiner: Tiku; Sisay G

Attorney, Agent or Firm: Honigman LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. national stage of PCT/JP2021/027304 filed on Jul. 21, 2021, which claims priority of Japanese Patent Application No. JP 2020-167159 filed on Oct. 1, 2020, the contents of which are incorporated herein.

TECHNICAL FIELD

(2) The present disclosure relates to a detection circuit and a power supply control device.

BACKGROUND

(3) JP 2019-106624A discloses a power supply system for a vehicle, which supplies power from a DC power source to a load. In this power supply system, a semiconductor switch is disposed on a current path of a current that flows from the positive electrode of the DC power source to an end of the load. The semiconductor switch is an N-channel FET (Field Effect Transistor). The positive electrode of the DC power source is connected to an input terminal of the semiconductor switch, namely the drain to which the current is input. A switching circuit steps up a voltage of the input terminal of the semiconductor switch and applies the stepped -up voltage to a control terminal of the semiconductor switch, namely the gate. As a result, the semiconductor switch is switched ON.

(4) In a conventional power supply system such as that described in JP 2019-106624A, the switching circuit steps up a voltage between the ground of the switching circuit and the input terminal of the semiconductor switch. The negative electrode of the DC power source is connected to a ground conductor, such as a vehicle body, to be grounded. The switching circuit is connected to the ground conductor via a connecting wire. While power is supplied to the switching circuit, a current flows successively through the switching circuit, the connecting wire, and the ground conductor. The ground potential of the switching circuit is the potential of an upstream end of the connecting wire.

(5) Let us assume that the resistance value of the connecting wire is not negligible. In this case, the voltage between the ground of the switching circuit and the input terminal of the semiconductor switch is lower than a voltage between the two terminals of the DC power source. The higher a voltage between two ends of the connecting wire is, the lower the voltage between the ground of the switching circuit and the input terminal of the semiconductor switch is. If the switching circuit steps up the voltage between the ground of the switching circuit and the input terminal of the semiconductor switch in a state where the voltage is lower than a certain value, an appropriate voltage may not be applied from the switching circuit to the semiconductor switch. In this case, a resistance value between the input terminal and an output terminal of the semiconductor switch, from which a current is output, may not decrease to a sufficiently small resistance value.

(6) If a current flows via the semiconductor switch in the state where the resistance value between the input terminal and the output terminal of the semiconductor switch is not sufficiently small, the semiconductor switch generates a large amount of heat. Therefore, the temperature of the semiconductor switch may rise to an abnormal temperature and the semiconductor switch may break down. In order to prevent breakdown of the semiconductor switch, the switching circuit is not instructed to switch the semiconductor switch ON if the voltage between the ground of the switching circuit and the input terminal of the semiconductor switch is lower than the certain value. In order to realize this configuration, it is necessary to detect not the voltage between the two terminals of the DC power source but the voltage between the ground of the switching circuit and the input terminal of the semiconductor switch.

SUMMARY

(7) Therefore, the present disclosure has an object of providing a detection circuit that detects a voltage between an input terminal of a semiconductor switch, to which a current is input, and the ground of a switching circuit that switches the semiconductor switch ON or OFF, and a power supply control device including the detection circuit.

(8) A detection circuit according to an aspect of the present disclosure includes: a current output device configured to output a current corresponding to a circuit voltage between an input terminal of a semiconductor switch to which a current is input and ground of a switching circuit configured to switch the semiconductor switch ON or OFF; and a circuit resistor through which the current output from the current output device flows, wherein a voltage between two ends of the circuit resistor is output, and a voltage between the ground of the switching circuit and a downstream end of the circuit resistor is higher than 0 V.

(9) A power supply control device according to an aspect of the present disclosure is configured to

control power that is supplied via a semiconductor switch, the power supply control device including: a switching circuit configured to switch the semiconductor switch ON or OFF; a detection circuit configured to detect a circuit voltage between an input terminal of the semiconductor switch to which a current is input and ground of the switching circuit, and output a voltage indicating the detected circuit voltage; and a processor configured to execute processing, wherein the processor instructs the switching circuit to switch the semiconductor switch ON or OFF in accordance with the voltage that is output from the detection circuit, the detection circuit includes: a current output device configured to output a current corresponding to the circuit voltage; and a circuit resistor through which the current output from the current output device flows, the detection circuit outputs a voltage between two ends of the circuit resistor, and a voltage between the ground of the switching circuit and a downstream end of the circuit resistor is higher than 0 V.

(10) Note that the present disclosure can be implemented not only as the power supply control device including the characteristic processor described above, but also as a power supply control method including a step of performing the characteristic processing described above, or a computer program for causing a computer to execute the step. Alternatively, the present disclosure can also be implemented as a semiconductor integrated circuit that implements some or all functions of the power supply control device, or a power supply system including the power supply control device.

(11) Effects of Present Disclosure

(12) According to the present disclosure, it is possible to detect a voltage between an input terminal of a semiconductor switch, to which a current is input, and the ground of a switching circuit that switches the semiconductor switch ON or OFF.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a block diagram showing the configuration of the main part of a power supply system according to a first embodiment.

(2) FIG. 2 is a circuit diagram of a first controller and a first detection circuit.

(3) FIG. 3 is a circuit diagram of a second controller and a second detection circuit.

(4) FIG. 4 is a block diagram showing the configuration of the main part of a microcomputer.

(5) FIG. 5 is a flowchart showing the procedure of second fault detection processing.

(6) FIG. 6 is a flowchart showing the procedure of second power supply control processing.

(7) FIG. 7 is a flowchart showing the procedure of second power supply control processing according to a second embodiment.

(8) FIG. 8 is a circuit diagram of a second detection circuit according to a third embodiment.

(9) FIG. 9 is a block diagram showing the configuration of the main part of a microcomputer.

(10) FIG. 10 is a flowchart showing the procedure of second fault detection processing.

(11) FIG. 11 is a flowchart showing the procedure of second power supply

(12) control processing.

(13) FIG. 12 is a flowchart showing the procedure of second power supply control processing according to a fourth embodiment.

(14) FIG. 13 is a block diagram showing the configuration of the main part of a power supply system according to a fifth embodiment.

(15) FIG. 14 is a block diagram showing the configuration of the main part of a power supply system according to a sixth embodiment.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(16) First, embodiments of the present disclosure will be described. At least some portions of the following embodiments may also be combined.

(17) First Aspect

(18) In accordance with a first aspect, a detection circuit according to an aspect of the present disclosure includes: a current output device configured to output a current corresponding to a circuit voltage between an input terminal of a semiconductor switch to which a current is input and ground of a switching circuit configured to switch the semiconductor switch ON or OFF; and a circuit resistor through which the current output from the current output device flows, wherein a voltage between two ends of the circuit resistor is output, and a voltage between the ground of the switching circuit and a downstream end of the circuit resistor is higher than 0 V.

(19) According to this aspect, a current corresponding to the circuit voltage is output in order to detect the circuit voltage. In a case where the resistance value of the circuit resistor is constant, the voltage between the two ends of the circuit resistor is proportional to a current output to the circuit resistor. The current corresponding to the circuit voltage flows via the circuit resistor. Therefore, the circuit voltage can be calculated based on the voltage between the two ends of the circuit resistor. Moreover, the circuit voltage is converted to the current corresponding to the circuit voltage, and therefore, the circuit voltage can be calculated by a microcomputer whose ground differs from the ground of the switching circuit.

(20) Second Aspect

(21) In a second aspect, in the detection circuit according to an aspect of the present disclosure, the current output device includes: a device resistor to which a current flows from the input terminal of the semiconductor switch; and a PNP bipolar transistor including an emitter connected to a downstream end of the device resistor, wherein a potential of a base of the bipolar transistor is a ground potential of the switching circuit, and a current output from a collector of the bipolar transistor flows to the circuit resistor.

(22) According to this aspect, the bipolar transistor adjusts a resistance value between the emitter and the collector in such a manner that a current proportional to the circuit voltage flows through the device resistor. Consequently, the circuit voltage is converted to the current proportional to the circuit voltage.

(23) Third Aspect

(24) In a third aspect, the detection circuit according to an aspect of the present disclosure includes a connection switch connected between the input terminal of the semiconductor switch and the current output device, wherein the current output device outputs a current drawn from the input terminal of the semiconductor switch to the circuit resistor.

(25) According to this aspect, the detection circuit operates when the connection switch is switched ON. The detection circuit stops operation when the connection switch is switched OFF. The detection circuit is caused to operate only when it is necessary to detect the circuit voltage. Therefore, unnecessary power consumption is suppressed.

(26) Fourth Aspect

(27) In a fourth aspect, a power supply control device according to an aspect of the present disclosure is configured to control power that is supplied via a semiconductor switch, the power supply control device including: a switching circuit configured to switch the semiconductor switch ON or OFF; a detection circuit configured to detect a circuit voltage between an input terminal of the semiconductor switch to which a current is input and ground of the switching circuit, and output a voltage indicating the detected circuit voltage; and a processor configured to execute processing, wherein the processor instructs the switching circuit to switch the semiconductor switch ON or OFF in accordance with the voltage that is output from the detection circuit, the detection circuit includes: a current output device configured to output a current corresponding to the circuit voltage; and a circuit resistor through which the current output from the current output device flows, the detection circuit outputs a voltage between two ends of the circuit resistor, and a voltage between the ground of the switching circuit and a downstream end of the circuit resistor is higher than 0 V.

(28) According to this aspect, the semiconductor switch is appropriately switched ON or OFF in accordance with the circuit voltage that is output from the detection circuit.

(29) Fifth Aspect

(30) In a fifth aspect, the power supply control device according to an aspect of the present disclosure further includes a current output unit configured to output a current corresponding to a current flowing through the semiconductor switch, and the voltage between the ground of the switching circuit and the downstream end of the circuit resistor fluctuates in accordance with the current output from the current output unit.

(31) According to this aspect, the circuit voltage fluctuates in accordance with the current output from the current output unit.

(32) Sixth Aspect

(33) In a sixth aspect, in the power supply control device according to an aspect of the present disclosure, the switching circuit steps up the circuit voltage, and applies the stepped-up voltage to a control terminal of the semiconductor switch to switch the semiconductor switch ON.

(34) According to this aspect, the semiconductor switch is an N-channel FET, for example. In this case, the input terminal and the control terminal of the semiconductor switch are the drain and the gate, respectively, for example. The switching circuit steps up the circuit voltage and applies the stepped-up voltage to the control terminal. As a result, the semiconductor switch is switched ON.

(35) Seventh Aspect

(36) In a seventh aspect, in the power supply control device according to an aspect of the present disclosure, the processor determines whether or not the circuit voltage is an ON voltage based on a voltage that is output from the detection circuit while the semiconductor switch is OFF, the ON voltage being a voltage that can be stepped up to a voltage with which the semiconductor switch can be switched ON, if it is determined that the circuit voltage is the ON voltage, the processor instructs the switching circuit to switch the semiconductor switch ON, and after giving the instruction to switch the semiconductor switch ON, the processor determines whether or not the semiconductor switch is ON.

(37) According to this aspect, whether or not the semiconductor switch can be switched ON is checked when the circuit voltage is the ON voltage.

(38) Eighth Aspect

(39) In an eighth aspect, in the power supply control device according to an aspect of the present disclosure, power is supplied to the switching circuit with a current that flows from the input terminal of the semiconductor switch successively through the switching circuit and the ground of the switching circuit, the processor determines whether or not the circuit voltage is lower than a threshold voltage based on a voltage that is output from the detection circuit while the semiconductor switch is OFF, and if it is determined that the circuit voltage is lower than the threshold voltage, the processor determines whether or not the switching circuit is operating.

(40) According to this aspect, the switching circuit stops operation when a voltage applied to the switching circuit is lower than a certain voltage. The threshold voltage is this voltage, for example. If the switching circuit is operating although a voltage between two ends of the switching circuit is lower than the threshold voltage, this means that a fault has occurred in the detection circuit. It is possible to detect a fault in the detection circuit by determining whether or not the switching circuit is operating.

(41) The following describes specific examples of power supply systems according to embodiments of the present disclosure with reference to the drawings. Note that the present disclosure is not limited to the following examples, but is defined by the claims, and is intended to encompass all modifications within the meanings and scope that are equivalent to the claims.

(42) First Embodiment

(43) Configuration of Power Supply System

(44) FIG. 1 is a block diagram showing the configuration of the main part of a power supply

system **1** according to a first embodiment. The power supply system **1** is mounted on a vehicle and includes a DC power source **10**, a first load **11**, a second load **12**, and a power supply control device **13**. The DC power source **10** is a battery, for example. A first lead wire **W1** and a second lead wire **W2** whose resistance values are not negligible are used for connections in the power supply system **1**. The equivalent circuits of the first lead wire **W1** and the second lead wire **W2** are represented by a first lead wire resistor **Rw1** and a second lead wire resistor **Rw2**, respectively. The first lead wire **W1** and the second lead wire **W2** are long lead wires, for example. If the first lead wire **W1** is long, the resistance value of the first lead wire resistor **Rw1** is not negligible. If the second lead wire **W2** is long, the resistance value of the second lead wire resistor **Rw2** is not negligible.

(45) The negative electrode of the DC power source **10** is connected to a ground conductor **G**. The negative electrode is grounded by being connected to the ground conductor **G**. The ground conductor **G** is a vehicle body, for example. The positive electrode of the DC power source **10** is connected to the power supply control device **13** and an end of the first lead wire resistor **Rw1**. The other end of the first lead wire resistor **Rw1** is connected to the power supply control device **13**. Furthermore, an end of the first load **11**, an end of the second load **12**, and an end of the second lead wire resistor **Rw2** are individually connected to the power supply control device **13**. The power supply control device **13**, the other end of the first load **11**, the other end of the second load **12**, and the other end of the second lead wire resistor **Rw2** are connected to the ground conductor **G**.

(46) The DC power source **10** supplies power to the power supply control device **13**. At this time, a current flows from the positive electrode of the DC power source **10** successively through the power supply control device **13** and the ground conductor **G**. The DC power source **10** supplies power to the first load **11** as well. At this time, a current flows from the positive electrode of the DC power source **10** successively through the power supply control device **13**, the first load **11**, and the ground conductor **G**. The DC power source **10** supplies power to the second load **12** as well. At this time, a current flows from the positive electrode of the DC power source **10** successively through the first lead wire resistor **Rw1**, the power supply control device **13**, the second load **12**, and the ground conductor **G**.

(47) The power supply control device **13** controls power supply from the DC power source **10** to the first load **11** and power supply from the DC power source **10** to the second load **12**. The first load **11** and the second load **12** are electric devices mounted on the vehicle. Each of the first load **11** and the second load **12** operates when power is supplied thereto. Each of the first load **11** and the second load **12** stops operation when power supply is stopped.

(48) Configuration of Power Supply Control Device **13**

(49) The power supply control device **13** includes a first controller **21**, a second controller **22**, a regulator **23**, a first detection circuit **24**, a second detection circuit **25**, and a microcomputer **26**. The first controller **21**, the regulator **23**, and the first detection circuit **24** included in the power supply control device **13** are connected to the positive electrode of the DC power source **10**. The first controller **21** is further connected to the upstream end of the first load **11** and the microcomputer **26**. Each of the regulator **23** and the first detection circuit **24** is further connected to the microcomputer **26**. The first controller **21**, the first detection circuit **24**, and the microcomputer **26** are connected to the ground conductor **G**.

(50) Each of the second controller **22** and the second detection circuit **25** is connected to the downstream end of the first lead wire resistor **Rw1**, the microcomputer **26**, and an end of the second lead wire resistor **Rw2**. The second controller **22** is further connected to the upstream end of the second load **12**. The second detection circuit **25** is further connected to the ground conductor **G**.

(51) A current flows from the positive electrode of the DC power source **10** successively through the first controller **21** and the ground conductor **G**, and the DC power source **10** supplies power to the first controller **21**. The first controller **21** includes a first FET **30** (see FIG. 2) that functions as a

semiconductor switch. The DC power source **10** supplies power to the first load **11** via the first FET **30**. The microcomputer **26** outputs a first instruction signal having a high-level voltage or a low-level voltage to the first controller **21**.

(52) The first FET **30** included in the first controller **21** is switched ON when the voltage of the first instruction signal is switched from the low-level voltage to the high-level voltage in a state where a first load current flowing through the first load **11** is less than a predetermined first threshold current. As a result, a current flows from the positive electrode of the DC power source **10** successively through the first FET **30**, the first load **11**, and the ground conductor G, and power is supplied to the first load **11**. When the voltage of the first instruction signal is switched from the high-level voltage to the low-level voltage or the first load current has risen to be greater than or equal to the first threshold current, the first FET **30** is switched OFF. As a result, power supply to the first load **11** stops and the first load **11** stops operation.

(53) In the following description, the voltage at the positive electrode of the DC power source **10** will be referred to as “power source voltage”. A reference potential of the power source voltage is the potential of the ground conductor G. The ground potential of the microcomputer **26** is the potential of the ground conductor G. The power source voltage fluctuates due to various factors. The first detection circuit **24** detects the power source voltage. The first detection circuit **24** outputs a voltage indicating the detected power source voltage as power source voltage information to the microcomputer **26**. The microcomputer **26** switches the voltage of the first instruction signal to the high-level voltage or the low-level voltage in accordance with the power source voltage indicated by the power source voltage information. Thus, the first FET **30** is switched in accordance with the power source voltage.

(54) The regulator **23** steps down the power source voltage to a predetermined voltage, and applies the stepped-down voltage to the microcomputer **26**. As a result, a current flows from the positive electrode of the DC power source **10** successively through the regulator **23**, the microcomputer **26**, and the ground conductor G, and power is supplied to the microcomputer **26**. A reference potential of the voltage stepped-down by the regulator **23** is the potential of the ground conductor G.

(55) A current flows from the positive electrode of the DC power source **10** successively through the first lead wire resistor **Rw1**, the second controller **22**, the second lead wire resistor **Rw2**, and the ground conductor G. The DC power source **10** supplies power to the second controller **22**. The second controller **22** includes a second FET **40** (see FIG. 3) that functions as a semiconductor switch. The DC power source **10** supplies power to the second load **12** via the second FET **40**. The microcomputer **26** outputs a second instruction signal having a high-level voltage or a low-level voltage to the second controller **22**.

(56) The second FET **40** is switched ON when the voltage of the second instruction signal is switched from the low-level voltage to the high-level voltage in a state where a second load current flowing through the second load **12** is less than a predetermined second threshold current. As a result, a current flows from the positive electrode of the DC power source **10** successively through the first lead wire resistor **Rw1**, the second FET **40**, the second load **12**, and the ground conductor G, and power is supplied to the second load **12**. When the voltage of the second instruction signal is switched from the high-level voltage to the low-level voltage or the second load current has risen to be greater than or equal to the second threshold current, the second FET **40** is switched OFF. As a result, power supply to the second load **12** stops and the second load **12** stops operation.

(57) As described above, the power supply control device **13** controls power supplied via the first FET **30** by switching the first FET **30** ON or OFF. Furthermore, the power supply control device **13** controls power supplied via the second FET **40** by switching the second FET **40** ON or OFF.

(58) The first FET **30** and the second FET **40** are N-channel FETs. In each of the first FET **30** and the second FET **40**, a current is input to the drain and is output from the source. In each of the first FET **30** and the second FET **40**, the drain functions as an input terminal to which a current is input, and the source functions as an output terminal from which the current is output.

(59) In the following description, the voltage at the drain of the second FET **40** will be referred to as “circuit voltage”. A reference potential of the circuit voltage is the potential at the upstream end of the second lead wire resistor **Rw2**. When the power source voltage fluctuates, the circuit voltage also fluctuates. The second detection circuit **25** detects the circuit voltage, and outputs a voltage indicating the detected circuit voltage as circuit voltage information to the microcomputer **26**. A reference potential of the circuit voltage information is the ground potential of the microcomputer **26**, namely the potential of the ground conductor **G**. The microcomputer **26** switches the voltage of the second instruction signal to the high -level voltage or the low-level voltage in accordance with the circuit voltage indicated by the circuit voltage information. Thus, the second FET **40** is switched in accordance with the circuit voltage.

(60) Configuration of First Controller **21**

(61) FIG. **2** is a circuit diagram of the first controller **21** and the first detection circuit **24**. The first controller **21** includes a first output unit **31**, a first drive circuit **32**, and a first voltage resistor **33** in addition to the first FET **30**. The drain of the first FET **30** is connected to the positive electrode of the DC power source **10**. The source of the first FET **30** is connected to the first output unit **31**. The first output unit **31** is further connected to the upstream end of the first load **11** and an end of the first voltage resistor **33**. The drain and the gate of the first FET **30** are individually connected to the first drive circuit **32**. The first drive circuit **32** is further connected to the microcomputer **26**. A connection node between the first output unit **31** and the first voltage resistor **33** is connected to the microcomputer **26** and the first drive circuit **32**. The first drive circuit **32** and the other end of the first voltage resistor **33** are connected to the ground conductor **G**.

(62) When the voltage at the gate of the first FET **30** relative to the potential of the source is higher than or equal to a predetermined first switch voltage, a resistance value between the drain and the source of the first FET **30** is sufficiently small. At this time, the first FET **30** is ON, and a current can flow through the drain and the source of the first FET **30**. When the first FET **30** is ON, a current flows from the positive electrode of the DC power source **10** successively through the first FET **30**, the first output unit **31**, the first load **11**, and the ground conductor **G**. When the voltage at the gate of the first FET **30** relative to the potential of the source is lower than the first switch voltage, a resistance value between the drain and the source of the first FET **30** is sufficiently large. At this time, the first FET **30** is OFF, and no current flows through the drain and the source of the first FET **30**.

(63) A current flows from the positive electrode of the DC power source **10** successively through the first drive circuit **32** and the ground conductor **G**, and power is supplied to the first drive circuit **32**. The first drive circuit **32** operates when the power source voltage has risen and become higher than or equal to a predetermined first operation voltage. When the power source voltage has dropped and become lower than the first operation voltage, the first drive circuit **32** stops operation. The ground potential of the first drive circuit **32** is the potential of the ground conductor **G**.

(64) The first drive circuit **32** steps up the power source voltage and applies the stepped-up voltage to the gate of the first FET **30**. As a result, the voltage at the gate of the first FET **30** relative to the potential of the source rises and becomes higher than or equal to the first switch voltage. As a result, the first FET **30** is switched ON. A reference potential of the stepped-up voltage is the potential of the ground conductor **G**. When the first drive circuit **32** stops stepping up the power source voltage, the voltage at the gate of the first FET **30** relative to the potential of the source drops to be lower than the first switch voltage. As a result, the first FET **30** is switched OFF. As described above, the first drive circuit **32** switches the first FET **30** ON or OFF by adjusting the voltage at the gate of the first FET **30**.

(65) The first output unit **31** is constituted by a current mirror circuit, for example, and outputs, to the first voltage resistor **33**, a current that is proportional to the first load current flowing through the first FET **30** to the first load **11**. The resistance value of the first voltage resistor **33** is constant. Therefore, the voltage between two ends of the first voltage resistor **33** is proportional to the

current flowing through the first voltage resistor **33**. Accordingly, the voltage between the two ends of the first voltage resistor **33** is proportional to the first load current. The voltage between the two ends of the first voltage resistor **33** is output as first current information indicating the first load current, to the microcomputer **26** and the first drive circuit **32**.

(66) As described above, the microcomputer **26** outputs a first instruction signal to the first drive circuit **32**. The first drive circuit **32** switches the first FET **30** ON when the voltage of the first instruction signal is switched from the low-level voltage to the high-level voltage in a state where the first load current indicated by the first current information is less than the first threshold current. The first drive circuit **32** switches the first FET **30** OFF when the voltage of the first instruction signal is switched from the high-level voltage to the low-level voltage or the first load current indicated by the first current information has risen to be greater than or equal to the first threshold current.

(67) The microcomputer **26** communicates with the first drive circuit **32**. The microcomputer **26** can determine whether or not the first drive circuit **32** is operating, by communicating with the first drive circuit **32**. For example, the microcomputer **26** transmits a request signal to the first drive circuit **32** to request transmission of a response signal. The microcomputer **26** determines whether or not the first drive circuit **32** is operating based on whether or not the response signal is received from the first drive circuit **32**.

(68) Configuration of First Detection Circuit **24**

(69) The first detection circuit **24** includes two voltage-dividing resistors **Rd1** and **Rd2**. An end of the voltage-dividing resistor **Rd1** is connected to the positive electrode of the DC power source **10**. The other end of the voltage-dividing resistor **Rd1** is connected to an end of the voltage-dividing resistor **Rd2**. The other end of the voltage-dividing resistor **Rd2** is connected to the ground conductor **G**. A connection node between the two voltage-dividing resistors **Rd1** and **Rd2** is connected to the microcomputer **26**.

(70) The first detection circuit **24** divides the power source voltage. The resistance values of the two voltage-dividing resistors **Rd1** and **Rd2** are constant. Accordingly, a voltage obtained as a result of the first detection circuit **24** dividing the power source voltage is proportional to the power source voltage. The first detection circuit **24** detects the power source voltage by dividing the power source voltage. The first detection circuit **24** outputs the voltage obtained by dividing the power source voltage as power source voltage information to the microcomputer **26**. As described above, the microcomputer **26** switches the voltage of the first instruction signal to the high-level voltage or the low-level voltage in accordance with the power source voltage indicated by the power source voltage information.

(71) Configuration of Second Controller **22**

(72) FIG. **3** is a circuit diagram of the second controller **22** and the second detection circuit **25**. The second controller **22** includes a second output unit **41**, a second drive circuit **42**, and a second voltage resistor **43** in addition to the second FET **40**. The drain of the second FET **40** is connected to the downstream end of the first lead wire resistor **Rw1**. The upstream end of the first lead wire resistor **Rw1** is connected to the positive electrode of the DC power source **10**. The source of the second FET **40** is connected to the second output unit **41**. The second output unit **41** is further connected to the upstream end of the second load **12** and an end of the second voltage resistor **43**. The drain and the gate of the second FET **40** are individually connected to the second drive circuit **42**. The second drive circuit **42** is further connected to the microcomputer **26**. A connection node between the second output unit **41** and the second voltage resistor **43** is connected to the microcomputer **26** and the second drive circuit **42**. The second drive circuit **42** and the other end of the second voltage resistor **43** are connected to the upstream end of the second lead wire resistor **Rw2**. As described above, the downstream end of the second lead wire resistor **Rw2** is connected to the ground conductor **G**.

(73) When the voltage at the gate of the second FET **40** relative to the potential of the source is

higher than or equal to a predetermined second switch voltage, a resistance value between the drain and the source of the second FET **40** is sufficiently small. At this time, the second FET **40** is ON, and a current can flow through the drain and the source of the second FET **40**. When the second FET **40** is ON, a current flows from the positive electrode of the DC power source **10** successively through the second FET **40**, the second output unit **41**, the second load **12**, and the ground conductor G. When the voltage at the gate of the second FET **40** relative to the potential of the source is lower than the second switch voltage, a resistance value between the drain and the source of the second FET **40** is sufficiently large. At this time, the second FET **40** is OFF, and no current flows through the drain and the source of the second FET **40**.

(74) A current flows from the positive electrode of the DC power source **10** successively through the first lead wire resistor **Rw1**, the second drive circuit **42**, the second lead wire resistor **Rw2**, and the ground conductor G. Thus, power is supplied to the second drive circuit **42**. The second drive circuit **42** operates when the circuit voltage has risen and become higher than or equal to a predetermined second operation voltage. The second drive circuit **42** stops operation when the circuit voltage has dropped and become lower than the second operation voltage. The ground potential of the second drive circuit **42** is the potential at the upstream end of the second lead wire resistor **Rw2**. While power is supplied to the second drive circuit **42**, a current flows from the drain of the second FET **40** successively through the second drive circuit **42** and the ground of the second drive circuit **42**.

(75) The second drive circuit **42** adjusts the voltage at the gate relative to the ground potential of the second drive circuit **42**. As described above, the reference potential of the circuit voltage is the potential at the upstream end of the second lead wire resistor **Rw2**, namely the ground potential of the second drive circuit **42**. The second drive circuit **42** steps up the circuit voltage and applies the stepped-up voltage to the gate of the second FET **40**. Consequently, the voltage at the gate of the second FET **40** relative to the potential of the source rises and becomes higher than or equal to the second switch voltage. As a result, the second FET **40** is switched ON. A reference potential of the stepped-up voltage is the ground potential of the second drive circuit **42**.

(76) The second drive circuit **42** stops stepping up the circuit voltage. As a result, the voltage at the gate relative to the ground potential of the second drive circuit **42** drops. Consequently, the voltage at the gate of the second FET **40** relative to the potential of the source drops to be lower than the second switch voltage. As a result, the second FET **40** is switched OFF.

(77) The second output unit **41** is constituted by a current mirror circuit, for example, and outputs a current that is proportional to the second load current flowing through the second FET **40** to the second load **12**. The second output unit **41** functions as a current output unit. The current output from the second output unit **41** flows via the second voltage resistor **43** and the second lead wire resistor **Rw2**. The resistance value of the second voltage resistor **43** is constant. Therefore, a voltage between two ends of the second voltage resistor **43** is proportional to a current flowing through the second voltage resistor **43**. Accordingly, the voltage between the two ends of the second voltage resistor **43** is proportional to the second load current. The voltage between the two ends of the second voltage resistor **43** is output as second current information indicating the second load current, to the second drive circuit **42**.

(78) The resistance value of the second lead wire resistor **Rw2** is also constant. Therefore, a voltage between two ends of a series circuit in which the second voltage resistor **43** and the second lead wire resistor **Rw2** are connected in series is proportional to a current flowing through the second voltage resistor **43**. Accordingly, the voltage between the two ends of the series circuit is proportional to the second load current. The voltage between the two ends of the series circuit is output as further second current information indicating the second load current, to the microcomputer **26**.

(79) As described above, the microcomputer **26** outputs a second instruction signal to the second drive circuit **42**. The second drive circuit **42** switches the second FET **40** ON when the voltage of

the second instruction signal is switched from the low-level voltage to the high-level voltage in a state where the second load current indicated by the second current information is less than the second threshold current. The second drive circuit **42** switches the second FET **40** OFF when the voltage of the second instruction signal is switched from the high-level voltage to the low-level voltage or the second load current indicated by the second current information has risen to be greater than or equal to the second threshold current.

(80) The microcomputer **26** communicates with the second drive circuit **42** as well as the first drive circuit **32**. The microcomputer **26** can determine whether or not the second drive circuit **42** is operating, by communicating with the second drive circuit **42**. For example, the microcomputer **26** transmits a request signal to the second drive circuit **42** to request transmission of a response signal, and determines whether or not the second drive circuit **42** is operating based on whether or not a response signal is received from the second drive circuit **42**.

(81) Configuration of Second Detection Circuit **25**

(82) The second detection circuit **25** includes a current output device **50** and a circuit resistor **51**. The current output device **50** includes a device resistor R_e and a transistor T_e . The transistor T_e is a PNP bipolar transistor.

(83) An end of the device resistor R_e is connected to the drain of the second FET **40**. The other end of the device resistor R_e is connected to the emitter of the transistor T_e . The base of the transistor T_e is connected to the upstream end of the second lead wire resistor R_{w2} . Accordingly, the potential of the base of the transistor T_e is the ground potential of the second drive circuit **42**. The collector of the transistor T_e is connected to the microcomputer **26** and an end of the circuit resistor **51**. The other end of the circuit resistor **51** is connected to the ground conductor G .

(84) A current flows from the positive electrode of the DC power source **10** successively through the first lead wire resistor R_{w1} , the device resistor R_e , the transistor T_e , the circuit resistor **51**, and the ground conductor G . Accordingly, the current flows to the device resistor R_e from the downstream end of the first lead wire resistor R_{w1} , namely the drain of the second FET **40**. In the transistor T_e , the current flows successively through the emitter and the collector. The emitter of the transistor T_e is connected to the downstream end of the device resistor R_e .

(85) A very small portion of the current input from the device resistor R_e to the emitter of the transistor T_e flows from the base via the second lead wire resistor R_{w2} to the ground conductor G . Substantially the total current input to the emitter is output from the collector. Therefore, the current output from the base is negligible, and the current input to the emitter substantially matches the current output from the collector.

(86) The voltage at the drain of the second FET **40** is denoted by V_d . The voltage at the upstream end of the second lead wire resistor R_{w2} is denoted by V_g .

(87) Reference potentials of the voltages V_d and V_g are the potential of the ground conductor G . $(V_d - V_g)$ represents the circuit voltage. The resistance value of the device resistor R_e is denoted by r_e . The voltage (an absolute value of the voltage) between the base and the emitter of the transistor T_e is denoted by V_f . A current flowing through the device resistor R_e is denoted by I_v .

(88) The transistor T_e adjusts a resistance value between the emitter and the collector in such a manner that the voltage at the emitter relative to the potential of the ground conductor G becomes $(V_g + V_f)$. Consequently, the transistor T_e adjusts the resistance value between the emitter and the collector in such a manner that the current I_v satisfies the following Equation (1).

(89)
$$I_v = (V_d - (V_g + V_f)) / r_e \quad (1)$$

(90) As described above, substantially the total current input to the emitter is output from the collector. Accordingly, the current flowing through the circuit resistor **51** substantially matches the current I_v .

(91) The following Equation (2) is obtained by developing the Equation (1).

(92)
$$I_v = ((V_d - V_g) - V_f) / r_e \quad (2)$$

(93) As described above, $(V_d - V_g)$ represents the circuit voltage. The voltage V_f and the resistance value r_e are substantially constant values, and therefore, the current I_v is proportional to the circuit voltage. Accordingly, the current output device **50** converts the circuit voltage to a current I_v that is proportional to the circuit voltage, and outputs the converted current I_v to the circuit resistor **51**. The current I_v is drawn from the drain of the second FET **40**. The current I_v output from the collector of the transistor T_e included in the current output device **50** flows via the circuit resistor **51**. The current output device **50** detects the circuit voltage by outputting the current I_v .

(94) The resistance value of the circuit resistor **51** is denoted by r_c . A voltage between two ends of the circuit resistor **51** is denoted by V_c . The voltage V_c is expressed by the following Equation (3).

(95) $V_c = r_c \cdot I_v$ (3)

(96) represents multiplication. When the current I_v is eliminated by substituting the Equation (2) into the Equation (3), the following Equation (4) is satisfied with respect to the voltage V_c .

(97) $V_c = ((V_d - V_g) - V_f) \cdot r_c / r_e$ (4)

(98) As described above, $(V_d - V_g)$ represents the circuit voltage. Not only the voltage V_f and the resistance value r_e but also the resistance value r_c is substantially constant. Therefore, the voltage V_c is proportional to the circuit voltage. Since the voltage V_f and the resistance values r_c and r_e are substantially constant, it is possible to calculate the circuit voltage, namely $(V_d - V_g)$ by substituting the voltage V_c into the Equation (4).

(99) As described above, the second detection circuit **25** detects the circuit voltage and outputs the voltage V_c , which is proportional to the detected circuit voltage, as circuit voltage information to the microcomputer **26**. The microcomputer **26** calculates the circuit voltage based on the circuit voltage information, namely the voltage V_c , and the Equation (4). The ground potential of the microcomputer **26** is the potential of the ground conductor G as described above, and differs from the ground potential of the second drive circuit **42**. However, the current output device **50** included in the second detection circuit **25** converts the circuit voltage to the current I_v , and therefore, the microcomputer **26** can calculate the circuit voltage.

(100) While power is supplied to the second drive circuit **42**, a current flows from the positive electrode of the DC power source **10** successively through the first lead wire resistor R_{w1} , the second drive circuit **42**, the second lead wire resistor R_{w2} , and the ground conductor G . The downstream end of the circuit resistor **51** is connected to the ground conductor G . As described above, the potential at the upstream end of the second lead wire resistor R_{w2} is the ground potential of the second drive circuit **42**. While power is supplied to the second drive circuit **42**, a current flows via the second lead wire resistor R_{w2} , and therefore, a voltage between the ground of the second drive circuit **42** and the downstream end of the circuit resistor **51** is higher than 0 V.

(101) As described above, the current output from the second output unit **41** is proportional to the second load current and flows via the second lead wire resistor R_{w2} . Therefore, when the second FET **40** is OFF, the current output from the second output unit **41** is 0 A. When the second FET **40** is switched ON, the current output from the second output unit **41** increases. As a result, the current flowing through the second lead wire resistor R_{w2} also increases. Also, when the second FET **40** is OFF, only a current that flows via the second drive circuit **42** flows through the first lead wire resistor R_{w1} , and this current is small. When the second FET **40** is switched ON, the sum of the current flowing through the second drive circuit **42** and the current that flows through the second FET **40** flows through the first lead wire resistor R_{w1} . For the reasons described above, the circuit voltage is higher when the second FET **40** is OFF than when the second FET **40** is ON.

(102) As described above, the current output from the second output unit **41** flows through the second lead wire resistor R_{w2} . Therefore, the voltage between the two ends of the second lead wire resistor R_{w2} , namely the voltage between the ground of the second drive circuit **42** and the downstream end of the circuit resistor **51** fluctuates in accordance with the current output from the second output unit **41**.

(103) Configuration of Microcomputer 26

(104) FIG. 4 is a block diagram showing the configuration of the main part of the microcomputer 26. The microcomputer 26 includes A/D converters 60, 61, 62, and 63, signal output units 64 and 65, communication units 66 and 67, a notification unit 68, a storage 69, and a controller 70. The A/D converters 60, 61, 62, and 63, the signal output units 64 and 65, the communication units 66 and 67, the notification unit 68, the storage 69, and the controller 70 are connected to an internal bus 71.

(105) The A/D converter 60 is further connected to the connection node between the voltage-dividing resistors Rd1 and Rd2 included in the first detection circuit 24. The signal output unit 64 and the communication unit 66 are further connected to the first drive circuit 32. The A/D converter 61 is further connected to the connection node between the first output unit 31 and the first voltage resistor 33 included in the first controller 21. The A/D converter 62 is further connected to the upstream end of the circuit resistor 51 included in the second detection circuit 25. The signal output unit 65 and the communication unit 67 are further connected to the second drive circuit 42. The A/D converter 63 is connected to the connection node between the second output unit 41 and the second voltage resistor 43 included in the second controller 22.

(106) Analog power source voltage information is input from the first detection circuit 24 to the A/D converter 60. The A/D converter 60 converts the input analog power source voltage information to digital power source voltage information. The controller 70 obtains the digital power source voltage information from the A/D converter 60.

(107) The signal output unit 64 outputs a first instruction signal to the first drive circuit 32. The signal output unit 64 switches the voltage of the first instruction signal to the high-level voltage or the low-level voltage in accordance with instructions from the controller 70.

(108) The communication unit 66 transmits a signal to the first drive circuit 32 in accordance with instructions from the controller 70. Furthermore, the communication unit 66 receives a signal from the first drive circuit 32.

(109) Analog first current information is input from the first controller 21 to the A/D converter 61. The A/D converter 61 converts the input analog first current information to digital first current information. The controller 70 obtains the digital first current information from the A/D converter 61.

(110) Analog circuit voltage information is input from the second detection circuit 25 to the A/D converter 62. The A/D converter 62 converts the input analog circuit voltage information to digital circuit voltage information. The controller 70 obtains the digital circuit voltage information from the A/D converter 62.

(111) The signal output unit 65 outputs a second instruction signal to the second drive circuit 42. The signal output unit 65 switches the voltage of the second instruction signal to the high-level voltage or the low-level voltage in accordance with instructions from the controller 70.

(112) The communication unit 67 transmits a signal to the second drive circuit 42 in accordance with instructions from the controller 70. Furthermore, the communication unit 67 receives a signal from the second drive circuit 42.

(113) Analog second current information is input from the second controller 22 to the A/D converter 63. The A/D converter 63 converts the input analog second current information to digital second current information. The controller 70 obtains the digital second current information from the A/D converter 63.

(114) The notification unit 68 gives notifications in accordance with instructions from the controller 70. The notifications are given by transmitting a signal, lighting a lamp, or displaying a message, for example.

(115) The storage 69 is a nonvolatile memory, for example. A computer program P is stored in the storage 69. The controller 70 includes a processing unit that executes processing, such as a CPU (Central Processing Unit). The processing unit of the controller 70 executes first fault detection

processing, first power supply control processing, second fault detection processing, second power supply control processing, and the like by executing the computer program P. The controller **70** functions as a processor.

(116) The first fault detection processing is processing for detecting a fault relating to the first controller **21** or the first detection circuit **24**. The first power supply control processing is processing for controlling power supply to the first load **11**. The second fault detection processing is processing for detecting a fault relating to the second controller **22** or the second detection circuit **25**. The second power supply control processing is processing for controlling power supply to the second load **12**.

(117) Note that the computer program P may also be stored in a storage medium A in such a manner that the processing unit of the controller **70** can read the program. In this case, the computer program P read out from the storage medium A by a reading device (not shown) is written into the storage **69**. Examples of the storage medium A include an optical disk, a flexible disk, a magnetic disk, a magneto-optical disk, and a semiconductor memory. Examples of the optical disk include a CD (Compact Disc)-ROM (Read Only Memory), a DVD (Digital Versatile Disc)-ROM, and a BD (Blu-ray (registered trademark)) Disc. An example of the magnetic disk is a hard disk. It is also possible to download the computer program P from a device (not shown) that is connected to a communication network (not shown), and write the downloaded computer program P into the storage **69**.

(118) The number of processing units included in the controller **70** is not limited to one, and may be two or more. In this case, a configuration is also possible in which the plurality of processing units execute the first fault detection processing, the first power supply control processing, the second fault detection processing, the second power supply control processing, and the like in cooperation with each other, following the computer program P.

(119) The technology of the present disclosure differs from conventional technologies mainly with regard to the second fault detection processing and the second power supply control processing. Therefore, the following first describes the second fault detection processing and the second power supply control processing. Thereafter, differences between the first fault detection processing and the second fault detection processing will be described to explain the first fault detection processing. Also, differences between the first power supply control processing and the second power supply control processing will be described to explain the first power supply control processing. These types of processing will be described on the assumption that the first load current does not rise to be greater than or equal to the first threshold current and the second load current does not rise to be greater than or equal to the second threshold current.

(120) Second Fault Detection Processing

(121) FIG. **5** is a flowchart showing the procedure of the second fault detection processing. The controller **70** executes the second fault detection processing before causing the second load **12** to operate. For example, the controller **70** executes the second fault detection processing when the engine of the vehicle is started while the second load **12** is not operating. Accordingly, the second fault detection processing is executed while the second FET **40** is OFF. If no fault is detected in the second fault detection processing, the controller **70** executes the second power supply control processing.

(122) In the second fault detection processing, the controller **70** obtains circuit voltage information from the A/D converter **62** (step **S1**). The circuit voltage information obtained in step **S1** is the voltage that is output from the second detection circuit **25** while the second FET **40** is OFF. The circuit voltage indicated by the circuit voltage information obtained in step **S1** is higher than the circuit voltage when the second FET **40** is ON. Next, the controller **70** determines whether or not the circuit voltage indicated by the circuit voltage information obtained in step **S1** is lower than the second operation voltage (step **S2**). The second drive circuit **42** ceases to operate when the circuit voltage is lower than the second operation voltage. The second operation voltage serves as a

threshold voltage.

(123) If it is determined that the circuit voltage indicated by the circuit voltage information is lower than the second operation voltage (S2: YES), the controller **70** determines whether or not the second drive circuit **42** is operating (step S3). If the second drive circuit **42** is operating when step S3 is executed, this means that the second detection circuit **25** did not output an appropriate voltage, namely a fault has occurred in the second detection circuit **25**. The controller **70** can detect a fault in the second detection circuit **25** by executing step S2.

(124) It is possible to determine whether or not the second drive circuit **42** is operating through communications, for example. In this method, the controller **70** causes the communication unit **66** to transmit a request signal to the second drive circuit **42** to request transmission of a response signal, for example. If the communication unit **66** receives a response signal in a predetermined period from when the request signal is transmitted, the controller **70** determines that the second drive circuit **42** is operating. If the communication unit **66** does not receive a response signal in the predetermined period from when the request signal is transmitted, the controller **70** determines that the second drive circuit **42** is not operating.

(125) Alternatively, it is possible to determine whether or not the second drive circuit **42** is operating by confirming the state of the second FET **40**. In this method, the controller **70** instructs the signal output unit **65** to switch the voltage of the second instruction signal from the low-level voltage to the high-level voltage, for example. The controller **70** obtains second current information from the A/D converter **63** while the second instruction signal has the high-level voltage, and then instructs the signal output unit **65** to return the voltage of the second instruction signal to the low-level voltage. If the second load current indicated by the second current information is greater than or equal to a predetermined value, this means that the second FET **40** is ON. If the second load current indicated by the second current information is less than the predetermined value, this means that the second FET **40** is OFF. If the second load current indicated by the second current information is greater than or equal to the predetermined value, the controller **70** determines that the second drive circuit **42** is operating. If the second load current indicated by the second current information is less than the predetermined value, the controller **70** determines that the second drive circuit **42** is not operating.

(126) If it is determined that the second drive circuit **42** is operating (S3: YES), the controller **70** determines that a fault has occurred in the second detection circuit **25** and keeps the second FET **40** ON or OFF (step S4). The controller **70** keeps the second FET **40** ON by instructing the signal output unit **65** to keep the voltage of the second instruction signal at the high-level voltage. The controller **70** keeps the second FET **40** OFF by instructing the signal output unit **65** to keep the voltage of the second instruction signal at the low-level voltage.

(127) The second FET is kept ON or OFF depending on the properties of the second load **12**. In the case where no problem occurs in driving even if the second load **12** stops operation, such as a case where the second load **12** is a room lamp, the controller **70** keeps the second FET **40** OFF in step S4. In cases where a problem may occur in driving if the second load **12** stops operation, such as a case where the second load **12** is a headlight, the controller **70** keeps the second FET **40** ON in step S4.

(128) If it is determined that the circuit voltage indicated by the circuit voltage information is higher than or equal to the second operation voltage (S2: NO), the controller **70** determines whether or not the circuit voltage indicated by the circuit voltage information obtained in step S1 is an ON voltage (step S5). If the circuit voltage is too low, the second drive circuit **42** cannot step up the circuit voltage to a voltage with which the second FET **40** can be switched ON. The ON voltage is a voltage that falls within a range set in advance and can be stepped up to a voltage with which the second FET **40** can be switched ON. If the circuit voltage is the ON voltage while the second FET **40** is OFF, the circuit voltage is high even when the second FET **40** is switched ON, and the second drive circuit **42** can switch the second FET **40** ON. If it is determined that the circuit voltage is the

ON voltage (S5: YES), the controller **70** instructs the signal output unit **65** to switch the second FET **40** ON (step S6). Specifically, the controller **70** instructs the signal output unit **65** to switch the voltage of the second instruction signal from the low-level voltage to the high-level voltage. As a result, the second drive circuit **42** switches the second FET **40** ON.

(129) Next, the controller **70** determines whether or not the second FET **40** is ON, based on a second load current indicated by second current information obtained from the A/D converter **63** (step S7). In step S7, if the second load current is greater than or equal to a predetermined value, the controller **70** determines that the second FET **40** is ON. If the second load current is less than the predetermined value, the controller **70** determines that the second FET **40** is not ON. The controller **70** confirms whether or not the second FET **40** can be switched ON, by executing steps S6 and S7.

(130) If it is determined that the second FET **40** is ON (S7: YES), the controller **70** instructs the signal output unit **65** to switch the second FET **40** OFF (step S8). Specifically, the controller **70** instructs the signal output unit **65** to switch the voltage of the second instruction signal to the low-level voltage. As a result, the second drive circuit **42** switches the second FET **40** OFF. After executing step S8, the controller **70** ends the second fault detection processing. Upon ending the second fault detection processing after executing step S8, the controller **70** executes the second power supply control processing.

(131) The controller **70** causes the notification unit **68** to give a notification (step S9) if it is determined that the second drive circuit is not operating (S3: NO), after executing step S4, if it is determined that the circuit voltage is not the ON voltage (S5: NO), or if it is determined that the second FET **40** is not ON (S7: NO). If it is determined that the second drive circuit is not operating or that the circuit voltage is not the ON voltage, the controller **70** instructs the notification unit **68** to output a signal indicating that the circuit voltage is low to a device (not shown), for example.

(132) If the notification is given after step S4 is executed, a signal indicating the occurrence of a fault in the second detection circuit **25** is output to the device, for example. If it is determined that the second FET **40** is not ON, the controller **70** instructs the notification unit **68** to output a signal indicating the occurrence of a fault in the second detection circuit **25** or the second drive circuit **42** to the device, for example.

(133) After executing step S9, the controller **70** ends the second fault detection processing. If the second fault detection processing is ended after step S9 is executed, the controller **70** does not execute the second power supply control processing.

(134) Second Power Supply Control Processing

(135) FIG. **6** is a flowchart showing the procedure of the second power supply control processing. The second power supply control processing is executed while the second FET **40** is OFF. In the second power supply control processing, the controller **70** determines whether or not to cause the second load **12** to operate (step S11). If a reception device (not shown) has received an instruction to operate the second load **12**, the controller **70** determines to cause the second load **12** to operate. If the reception device has not received an instruction to operate the second load **12**, the controller **70** determines not to cause the second load **12** to operate.

(136) If it is determined not to cause the second load **12** to operate (S11: NO), the controller **70** ends the second power supply control processing and again starts the second power supply control processing. Therefore, when determining not to cause the second load **12** to operate in step S11, the controller **70** again executes step S11.

(137) If it is determined to cause the second load **12** to operate (S11: YES), the controller **70** obtains circuit voltage information from the A/D converter **62** (step S12). The circuit voltage indicated by the circuit voltage information obtained in step S12 is the circuit voltage when the second FET **40** is OFF. Next, the controller **70** determines whether or not the circuit voltage indicated by the circuit voltage information obtained in step S12 falls within a normal range (step S13). The normal range is a range of the circuit voltage that enables the second drive circuit **42** to

appropriately switch the second FET **40** ON or OFF.

(138) If it is determined that the circuit voltage falls within the normal range (S13: YES), the controller **70** instructs the signal output unit **65** to switch the second FET **40** ON (step S14) as in step S6 of the second fault detection processing. As a result, power is supplied to the second load **12** and the second load **12** operates. After executing step S14, the controller **70** obtains circuit voltage information from the A/D converter **62** (step S15). The circuit voltage indicated by the circuit voltage information obtained in step S15 is the circuit voltage when the second FET **40** is ON.

(139) Next, the controller **70** determines whether or not the circuit voltage indicated by the circuit voltage information obtained in step S15 falls within the normal range (step S16). If it is determined that the circuit voltage falls within the normal range (S16: YES), the controller **70** determines whether or not to stop operation of the second load **12** (step S17). For example, if the reception device has received an instruction to stop operation of the second load **12**, the controller **70** determines to stop operation of the second load **12**. If the reception device has not received an instruction to stop operation of the second load **12**, the controller **70** determines not to stop operation of the second load **12**.

(140) If it is determined not to stop operation of the second load **12** (S17: NO), the controller **70** again executes step S15, and executes steps S16 and S17 repeatedly until the circuit voltage changes to a voltage that does not fall within the normal range or operation of the second load **12** is stopped. If it is determined that the circuit voltage does not fall within the normal range (S16: NO) or if it is determined to stop operation of the second load **12** (S17: YES), the controller **70** instructs the signal output unit **65** to switch the second FET **40** OFF (step S18) as in step S8 of the second fault detection processing, and ends the second power supply control processing. After ending the second power supply control processing, the controller **70** again executes the second power supply control processing. The controller **70** stops repeating the second power supply control processing when the engine is stopped, for example.

(141) As described above, in the second power supply control processing, the controller **70** instructs the signal output unit **65** to switch the second FET **40** ON or OFF in accordance with a circuit voltage indicated by circuit voltage information output from the second detection circuit **25**. The second drive circuit **42** switches the second FET **40** ON or OFF following the instruction from the controller **70**. Therefore, the second FET **40** is appropriately switched ON or OFF by the second drive circuit **42** in accordance with the circuit voltage detected by the second detection circuit **25**. Power supply to the second load **12** is appropriately controlled. The second drive circuit **42** functions as a switching circuit.

(142) First Fault Detection Processing and First Power Supply Control Processing

(143) The controller **70** executes the first fault detection processing similarly to the second fault detection processing. The controller **70** executes the first power supply control processing similarly to the second power supply control processing. The second FET **40**, the second drive circuit **42**, the circuit voltage information, the second operation voltage, and the second current information referred to in the descriptions of the second fault detection processing and the second power supply control processing respectively correspond to the first FET **30**, the first drive circuit **32**, the power source voltage information, the first operation voltage, and the first current information in the first fault detection processing and the first power supply control processing.

(144) Also, the A/D converters **62** and **63**, the signal output unit **65**, and the communication unit **67** referred to in the descriptions of the second fault detection processing and the second power supply control processing respectively correspond to the A/D converters **60** and **61**, the signal output unit **64**, and the communication unit **66** in the first fault detection processing and the first power supply control processing. A normal range in the first power supply control processing is a range of the power source voltage that enables the first drive circuit **32** to appropriately switch the first FET **30** ON or OFF.

(145) Second Embodiment

(146) In the second power supply control processing according to the first embodiment, the second drive circuit **42** switches the second FET **40** ON or OFF based on whether or not the circuit voltage falls within the normal range. However, there is no limitation to a configuration in which the second FET **40** is switched based on whether or not the circuit voltage falls within the normal range.

(147) The following describes differences of a second embodiment from the first embodiment. Configurations other than those described below are the same as those in the first embodiment. Therefore, structural features that are the same as those in the first embodiment are denoted by the same reference signs as those used in the first embodiment, and descriptions of those structural features are omitted.

(148) Configuration of Microcomputer **26**

(149) In the second embodiment, the signal output unit **65** shown in FIG. **4** outputs not the second instruction signal but a second PWM signal to the second drive circuit **42**. Similarly to the second instruction signal, the second PWM signal has a high-level voltage or a low-level voltage.

Similarly to the first embodiment, the second drive circuit **42** switches the second FET **40** ON when the voltage of the second PWM signal is switched from the low-level voltage to the high-level voltage. The second drive circuit **42** switches the second FET **40** OFF when the voltage of the second PWM signal is switched from the high-level voltage to the low-level voltage. The second drive circuit **42** keeps the second FET **40** OFF while output of the second PWM signal is stopped.

(150) The voltage of the second PWM signal is periodically switched from the low-level voltage to the high-level voltage. The duty cycle of the second PWM signal is adjusted by adjusting timings at which the high-level voltage is switched to the low-level voltage. The duty cycle is the ratio of a time for which the second PWM signal has the high-level voltage in a period.

(151) Note that a configuration is also possible in which the voltage of the second PWM signal is periodically switched from the high-level voltage to the low-level voltage. In this case, the duty cycle of the second PWM signal is adjusted by adjusting timings at which the low-level voltage is switched to the high-level voltage.

(152) The signal output unit **65** outputs the second PWM signal and stops outputting the second PWM signal in accordance with instructions from the controller **70**. The controller **70** sets the duty cycle of the second PWM signal in the signal output unit **65**. While outputting the second PWM signal, the signal output unit **65** adjusts the duty cycle of the second PWM signal to the set value.

(153) Second Power Supply Control Processing

(154) FIG. **7** is a flowchart showing the procedure of second power supply control processing according to the second embodiment. In the second power supply control processing according to the second embodiment, steps that are the same as those in the second power supply control processing according to the first embodiment are denoted by the same reference signs as those used in the first embodiment, and descriptions of those steps are omitted. The steps common to the first and second embodiments are steps **S11**, **S12**, **S15**, and **S17**.

(155) Similarly to the first embodiment, the second power supply control processing according to the second embodiment is executed while the second FET **40** is OFF. In the second power supply control processing, the controller **70** first executes step **S11**. If it is determined not to cause the second load **12** to operate (**S11**: NO), the controller **70** ends the second power supply control processing and again starts the second power supply control processing. Accordingly, when determining not to cause the second load **12** to operate in step **S11**, the controller **70** again executes step **S11**.

(156) If it is determined to cause the second load **12** to operate (**S11**: YES), the controller **70** executes step **S12**. After obtaining circuit voltage information from the A/D converter **62** in step **S12**, the controller **70** sets the duty cycle of the second PWM signal in the signal output unit **65** in accordance with a circuit voltage indicated by the obtained circuit voltage information (step **S21**).

The circuit voltage indicated by the circuit voltage information obtained in step S12 is a circuit voltage of the case where the second FET **40** is OFF.

(157) Next, the controller **70** instructs the signal output unit **65** to output the second PWM signal to the second drive circuit **42** (step S22). As a result, the second drive circuit **42** repeatedly switches the second FET **40** between ON and OFF in accordance with the voltage of the second PWM signal. Consequently, a current flows successively through the first lead wire resistor **Rw1**, the second FET **40**, the second output unit **41**, the second load **12**, and the ground conductor **G**, and power is supplied to the second load **12**. The average value of voltage applied to the second load **12** is adjusted in accordance with the duty cycle of the second PWM signal.

(158) After executing step S22, the controller **70** executes step S15. In step S15, the controller **70** obtains circuit voltage information while the second FET **40** is ON, namely the second PWM signal has the high-level voltage. After obtaining the circuit voltage information in step S15, the controller **70** changes the duty cycle of the second PWM signal in accordance with a circuit voltage indicated by the obtained circuit voltage information (step S23). Thus, the duty cycle of the second PWM signal is changed. After executing step S23, the controller **70** executes step S17.

(159) If it is determined not to stop operation of the second load **12** (S17: NO), the controller **70** executes step S15. The controller **70** repeatedly changes the duty cycle in accordance with the circuit voltage until the controller **70** determines to stop operation of the second load **12**. If it is determined to stop operation of the second load **12** (S17: YES), the controller **70** instructs the signal output unit **65** to stop outputting the second PWM signal (step S24), and ends the second power supply control processing. After ending the second power supply control processing, the controller **70** again executes the second power supply control processing. The controller **70** stops repeating the second power supply control processing when the engine is stopped, for example.

(160) As described above, in the second power supply control processing, the controller **70** changes the duty cycle of the second PWM signal in accordance with a circuit voltage indicated by circuit voltage information output from the second detection circuit **25**. The second drive circuit **42** switches the second FET **40** ON or OFF in accordance with the voltage of the second PWM signal. Therefore, the average value of voltage applied to the second load **12** is adjusted to an appropriate value, and power supply to the second load **12** is appropriately controlled. The second load **12** in the second embodiment is a lamp, for example.

(161) The effects obtained with use of the power supply control device **13** and the second detection circuit **25** according to the first embodiment are respectively obtained with use of the power supply control device **13** and the second detection circuit **25** according to the second embodiment.

(162) Supplementary Note

(163) In the second embodiment, the signal output unit **64** may also be configured similarly to the signal output unit **65**. In this case, the signal output unit **64** outputs a first PWM signal similar to the second PWM signal, and the first drive circuit **32** switches the first FET **30** ON or OFF in accordance with a voltage of the first PWM signal. The controller **70** executes first power supply control processing similarly to the second power supply control processing.

(164) The second FET **40**, the second drive circuit **42**, and the circuit voltage information referred to in the description of the second power supply control processing respectively correspond to the first FET **30**, the first drive circuit **32**, and the power source voltage information in the first power supply control processing. The A/D converters **62** and **63** and the signal output unit **65** referred to in the description of the second power supply control processing respectively correspond to the A/D converters **60** and **61** and the signal output unit **64** in the first power supply control processing.

(165) Third Embodiment

(166) In the first embodiment, a current always flows through the second detection circuit **25** and the second detection circuit **25** always detects the circuit voltage. However, a configuration is also possible in which the second detection circuit **25** does not always detect the circuit voltage.

(167) The following describes differences of a third embodiment from the first embodiment.

Configurations other than those described below are the same as those in the first embodiment. Therefore, structural features that are the same as those in the first embodiment are denoted by the same reference signs as those used in the first embodiment, and descriptions of those structural features are omitted.

(168) Configuration of Second Detection Circuit 25

(169) FIG. 8 is a circuit diagram of a second detection circuit 25 according to the third embodiment. Similarly to the first embodiment, the second detection circuit 25 according to the third embodiment includes a current output device 50 and a circuit resistor 51. The current output device 50 and the circuit resistor 51 are connected in the same manner as in the first embodiment, except for the upstream end of the device resistor Re.

(170) The second detection circuit 25 according to the third embodiment further includes a first circuit switch 52, a second circuit switch 53, and switch resistors 54, 55, 56, and 57. The first circuit switch 52 is a PNP bipolar transistor. The second circuit switch 53 is an NPN bipolar transistor.

(171) The emitter of the first circuit switch 52 is connected to the drain of the second FET 40. The collector of the first circuit switch 52 is connected to the upstream end of the device resistor Re. The switch resistor 54 is connected between the emitter and the base of the first circuit switch 52. An end of the switch resistor 55 is connected to the base of the first circuit switch 52. The first circuit switch 52 functions as a connection switch.

(172) The other end of the switch resistor 55 is connected to the collector of the second circuit switch 53. The emitter of the second circuit switch 53 is connected to the ground conductor G. The switch resistor 56 is connected between the emitter and the base of the second circuit switch 53. The base of the second circuit switch 53 is connected to the microcomputer 26 via the switch resistor 57.

(173) The microcomputer 26 adjusts the voltage of the base of the second circuit switch 53 relative to the potential of the ground conductor G to a high-level voltage or a low-level voltage. The second circuit switch 53 is switched ON when the voltage of the base of the second circuit switch 53 is adjusted to the high-level voltage by the microcomputer 26. When the second circuit switch 53 is ON, a resistance value between the collector and the emitter of the second circuit switch 53 is sufficiently small, and a current can flow via the collector and the emitter.

(174) When the second circuit switch 53 is ON, a current flows from the upstream end of the second controller 22 successively through the switch resistors 54 and 55, the second circuit switch 53, and the ground conductor G. As a result, the first circuit switch 52 is switched ON. When the first circuit switch 52 is ON, a resistance value between the emitter and the collector of the first circuit switch 52 is sufficiently small, and a current can flow via the emitter and the collector. When the first circuit switch 52 is ON, the current output device 50 draws a current from the drain of the second FET 40 via the first circuit switch 52, and the second detection circuit 25 detects the circuit voltage. As described above, the second detection circuit 25 operates when the first circuit switch 52 is ON.

(175) The second circuit switch 53 is switched OFF when the voltage of the base of the second circuit switch 53 is adjusted to the low-level voltage by the microcomputer 26. When the second circuit switch 53 is OFF, the resistance value between the collector and the emitter of the second circuit switch 53 is sufficiently large, and no current flows through the collector and the emitter.

(176) When the second circuit switch 53 is switched OFF, the flow of a current via the switch resistor 54 stops, and accordingly, the first circuit switch 52 is switched OFF. When the first circuit switch 52 is OFF, the resistance value between the emitter and the collector of the first circuit switch 52 is sufficiently large, and no current flows through the emitter and the collector. When the first circuit switch 52 is switched OFF, the current output device 50 stops drawing a current via the first circuit switch 52, and the second detection circuit 25 stops operation.

(177) As described above, the microcomputer 26 switches the first circuit switch 52 ON or OFF by

adjusting the voltage of the base of the second circuit switch **53** to the high-level voltage or the low-level voltage.

(178) Configuration of First Detection Circuit **24**

(179) Similarly to the second detection circuit **25**, a first detection circuit **24** according to the third embodiment includes a first circuit switch **52**, a second circuit switch **53**, and switch resistors **54**, **55**, **56**, and **57** in addition to the voltage-dividing resistors **Rd1** and **Rd2**. The first circuit switch **52**, the second circuit switch **53**, and the switch resistors **54**, **55**, **56**, and **57** are connected in the same manner as in the second detection circuit **25**. Accordingly, the first circuit switch **52** is connected between the first controller **21** and the upstream end of the voltage-dividing resistor **Rd1**. The emitter of the second circuit switch **53** is connected to the downstream end of the voltage-dividing resistor **Rd2**. The base of the second circuit switch **53** is connected to the microcomputer **26** via the switch resistor **57**.

(180) The microcomputer **26** switches the first circuit switch **52** ON or OFF by adjusting the voltage of the base of the second circuit switch **53** relative to the potential of the ground conductor **G** to the high-level voltage or the low-level voltage. When the voltage of the base of the second circuit switch **53** is switched to the high-level voltage, the first circuit switch **52** is switched ON, and the first detection circuit **24** operates. When the voltage of the base of the second circuit switch **53** is switched to the low-level voltage, the first circuit switch **52** is switched OFF, and the first detection circuit **24** stops operation.

(181) Configuration of Microcomputer **26**

(182) FIG. **9** is a block diagram showing the configuration of the main part of a microcomputer **26**. The microcomputer **26** according to the third embodiment includes switching units **72** and **73** in addition to the constitutional units included in the microcomputer **26** according to the first embodiment. The switching units **72** and **73** are connected to the internal bus **71**. The switching unit **72** is further connected to the second circuit switch **53** via the switch resistor **57** in the first detection circuit **24**. The switching unit **73** is further connected to the second circuit switch **53** via the switch resistor **57** in the second detection circuit **25**.

(183) The switching unit **72** adjusts the voltage of the base of the second circuit switch **53** included in the first detection circuit **24** relative to the potential of the ground conductor **G** to the high-level voltage or the low-level voltage. The switching unit **72** switches the first circuit switch **52** ON or OFF in accordance with instructions from the controller **70**. Specifically, the switching unit **72** switches the second circuit switch **53** ON by adjusting the voltage at the base of the second circuit switch **53** to the high-level voltage. As a result, the first circuit switch **52** is switched ON. The switching unit **72** switches the second circuit switch **53** OFF by adjusting the voltage at the base of the second circuit switch **53** to the low-level voltage. As a result, the first circuit switch **52** is switched OFF.

(184) Likewise, the switching unit **73** adjusts the voltage at the base of the second circuit switch **53** included in the second detection circuit **25** relative to the potential of the ground conductor **G** to the high-level voltage or the low-level voltage. The switching unit **73** switches the first circuit switch **52** ON or OFF in accordance with instructions from the controller **70**. Specifically, the switching unit **73** switches the second circuit switch **53** ON by adjusting the voltage at the base of the second circuit switch **53** to the high-level voltage. As a result, the first circuit switch **52** is switched ON. The switching unit **73** switches the second circuit switch **53** OFF by adjusting the voltage at the base of the second circuit switch **53** to the low-level voltage. As a result, the first circuit switch **52** is switched OFF.

(185) Second Fault Detection Processing

(186) FIG. **10** is a flowchart showing the procedure of second fault detection processing. Similarly to the first embodiment, the controller **70** executes the second fault detection processing before causing the second load **12** to operate. Accordingly, the second fault detection processing is executed while the second FET **40** is OFF. If no fault is detected in the second fault detection

processing, the controller **70** executes second power supply control processing.

(187) All the steps executed in the second fault detection processing according to the first embodiment are executed in the second fault detection processing according to the third embodiment. These steps are denoted by the same reference signs as those used in the first embodiment, and descriptions thereof are omitted. The steps executed in the second fault detection processing according to the first embodiment are steps **S1** to **S9**.

(188) In the second fault detection processing, the controller **70** first causes the second detection circuit **25** to operate by instructing the switching unit **72** to switch ON the first circuit switch **52** included in the second detection circuit **25** (step **S31**).

(189) After executing step **S31**, the controller **70** executes step **S1** to obtain circuit voltage information from the A/D converter **62**.

(190) After executing step **S1**, the controller **70** stops operation of the second detection circuit **25** by instructing the switching unit **72** to switch OFF the first circuit switch **52** included in the second detection circuit **25** (step **S32**). After executing step **S32**, the controller **70** executes step **S2**.

(191) Second Power Supply Control Processing

(192) FIG. **11** is a flowchart showing the procedure of the second power supply control processing. Similarly to the first embodiment, the second power supply control processing is executed while the second FET **40** is OFF.

(193) All the steps executed in the second power supply control processing according to the first embodiment are executed in the second power supply control processing according to the third embodiment. These steps are denoted by the same reference signs as those used in the first embodiment, and descriptions thereof are omitted. The steps executed in the second power supply control processing according to the first embodiment are steps **S11** to **S18**.

(194) In the second power supply control processing, if it is determined to cause the second load **12** to operate (**S11**: YES), the controller **70** causes the second detection circuit **25** to operate by instructing the switching unit **72** to switch ON the first circuit switch **52** included in the second detection circuit **25** (step **S41**). After executing step **S41**, the controller **70** executes step **S12**.

(195) After executing step **S18** or if it is determined that the circuit voltage indicated by circuit voltage information obtained in step **S12** does not fall within the normal range (**S13**: NO), the controller **70** stops operation of the second detection circuit **25** by instructing the switching unit **72** to switch OFF the first circuit switch **52** included in the second detection circuit **25** (step **S42**). After executing step **S42**, the controller **70** ends the second power supply control processing.

(196) First Fault Detection Processing and First Power Supply Control Processing

(197) The controller **70** executes first fault detection processing similarly to the second fault detection processing. The controller **70** executes first power supply control processing similarly to the second power supply control processing. When it is necessary to obtain power source voltage information, the controller **70** instructs the switching unit **72** to switch ON the first circuit switch **52** included in the first detection circuit **24**. As a result, the first detection circuit **24** operates. When it is no longer necessary to obtain power source voltage information, the controller **70** instructs the switching unit **72** to switch OFF the first circuit switch **52** included in the first detection circuit **24**. As a result, the first detection circuit **24** stops operation.

(198) As described above, in the power supply control device **13** according to the third embodiment, the first detection circuit **24** is caused to operate only when it is necessary to detect the power source voltage. The second detection circuit **25** is caused to operate only when it is necessary to detect the circuit voltage. Therefore, unnecessary power consumption is suppressed.

(199) The effects obtained with use of the power supply control device **13** according to the first embodiment are obtained with use of the power supply control device **13** according to the third embodiment.

(200) Fourth Embodiment

(201) The microcomputer **26**, the first detection circuit **24**, and the second detection circuit **25** in

the second embodiment may also be configured in the same manner as in the third embodiment.
(202) The following describes differences of a fourth embodiment from the second embodiment. Configurations other than those described below are the same as those in the second embodiment. Therefore, structural features that are the same as those in the second embodiment are denoted by the same reference signs as those used in the second embodiment, and descriptions of those structural features are omitted.

(203) First Fault Detection Processing and Second Fault Detection Processing

(204) In a power supply control device **13** according to the fourth embodiment, the microcomputer **26**, the first detection circuit **24**, and the second detection circuit **25** are configured in the same manner as in the third embodiment. The controller **70** included in the microcomputer **26** executes first fault detection processing and second fault detection processing in the same manner as in the third embodiment.

(205) Second Power Supply Control Processing

(206) FIG. **12** is a flowchart showing the procedure of second power supply control processing according to the fourth embodiment. Similarly to the second embodiment, the second power supply control processing is executed while the second FET **40** is OFF.

(207) All the steps executed in the second power supply control processing according to the second embodiment are executed in the second power supply control processing according to the fourth embodiment. These steps are denoted by the same reference signs as those used in the second embodiment, and descriptions thereof are omitted. The steps executed in the second power supply control processing according to the second embodiment are steps **S11**, **S12**, **S15**, **S17**, and **S21** to **S24**.

(208) In the second power supply control processing, if it is determined to cause the second load **12** to operate (**S11**: YES), the controller **70** causes the second detection circuit **25** to operate by instructing the switching unit **72** to switch ON the first circuit switch **52** included in the second detection circuit **25** (step **S51**). After executing step **S51**, the controller **70** executes step **S12**.

(209) After executing step **S24**, the controller **70** stops operation of the second detection circuit **25** by instructing the switching unit **72** to switch OFF the first circuit switch **52** included in the second detection circuit **25** (step **S52**). After executing step **S52**, the controller **70** ends the second power supply control processing.

(210) First Power Supply Control Processing

(211) The controller **70** executes first power supply control processing similarly to the second power supply control processing. The controller **70** causes the first detection circuit **24** to operate when it is necessary to obtain power source voltage information. The controller **70** stops operation of the first detection circuit **24** when it is no longer necessary to obtain power source voltage information.

(212) As described above, in the power supply control device **13** according to the fourth embodiment, the first detection circuit **24** is caused to operate only when it is necessary to detect the power source voltage. The second detection circuit **25** is caused to operate only when it is necessary to detect the circuit voltage. Therefore, unnecessary power consumption is suppressed.

(213) The effects obtained with use of the power supply control device **13** according to the second embodiment are obtained with use of the power supply control device **13** according to the fourth embodiment.

(214) Supplementary Note

(215) In the third and fourth embodiments, the first circuit switch **52** is not limited to the PNP bipolar transistor, and may also be a P-channel FET. The second circuit switch **53** is not limited to the NPN bipolar transistor, and may also be an N-channel FET. Also, the first circuit switch **52** is not limited to the semiconductor switch, and may also be a relay contact. The circuit for switching the first circuit switch **52** ON or OFF is not limited to the circuit in which the second circuit switch **53** is used.

(216) Fifth Embodiment

(217) The number of DC power sources included in the power supply system **1** according to the first embodiment is one. However, the number of DC power sources included in the power supply system **1** is not limited to one.

(218) The following describes differences of a fifth embodiment from the first embodiment.

Configurations other than those described below are the same as those in the first embodiment.

Therefore, structural features that are the same as those in the first embodiment are denoted by the same reference signs as those used in the first embodiment, and descriptions of those structural features are omitted.

(219) Configuration of Power Supply System **1**

(220) FIG. **13** is a block diagram showing the configuration of the main part of a power supply system **1** according to the fifth embodiment. The power supply system **1** according to the fifth embodiment includes a second DC power source **14** in addition to the structural features included in the power supply system **1** according to the first embodiment. The second DC power source **14** is a battery, for example. The negative electrode of the second DC power source **14** is connected to the ground conductor G. The positive electrode of the second DC power source **14** is connected to an end of the first lead wire resistor Rw**1**. The other end of the first lead wire resistor Rw**1** is connected in the same manner as in the first embodiment. In the fifth embodiment, not the DC power source **10** but the second DC power source **14** functions as a power source of the second load **12**, the second controller **22**, and the second detection circuit **25**.

(221) The effects obtained with use of the power supply control device **13**, the first detection circuit **24**, and the second detection circuit **25** according to the first embodiment are respectively obtained with use of the power supply control device **13**, the first detection circuit **24**, and the second detection circuit **25** according to the fifth embodiment.

(222) Sixth Embodiment

(223) The downstream end of the second load **12** is connected to the ground conductor G in the first embodiment. However, there is no limitation to the configuration in which the downstream end of the second load **12** is connected to the ground conductor G. The following describes differences of a sixth embodiment from the first embodiment. Configurations other than those described below are the same as those in the first embodiment. Therefore, structural features that are the same as those in the first embodiment are denoted by the same reference signs as those used in the first embodiment, and descriptions of those structural features are omitted.

(224) Configuration of Power Supply System **1**

(225) FIG. **14** is a block diagram showing the configuration of the main part of a power supply system **1** according to the sixth embodiment. In the power supply system **1** according to the sixth embodiment, the downstream end of the second load **12** is connected to the upstream end of the second lead wire resistor Rw**2**. In the power supply system **1** according to the sixth embodiment, when the second FET **40** included in the second controller **22** is ON, a current flows from the positive electrode of the DC power source **10** successively through the first lead wire resistor Rw**1**, the second FET **40**, the second output unit **41**, the second load **12**, the second lead wire resistor Rw**2**, and the ground conductor G. In this case, the circuit voltage matches the voltage at the drain of the second FET **40** relative to the potential of the downstream end of the second load **12**.

(226) The effects obtained with use of the power supply control device **13**, the first detection circuit **24**, and the second detection circuit **25** according to the first embodiment are respectively obtained with use of the power supply control device **13**, the first detection circuit **24**, and the second detection circuit **25** according to the sixth embodiment.

(227) Supplementary Note

(228) In the second to fourth and sixth embodiments, the power supply system **1** may also include the DC power source **10** and the second DC power source **14** similarly to the fifth embodiment.

(229) In the second to fifth embodiments, the downstream end of the second load **12** may also be

connected to the upstream end of the second lead wire resistor **Rw2** similarly to the sixth embodiment.

(230) As long as the second lead wire **W2** is used in the first to sixth embodiments, it is necessary to detect the circuit voltage, and the second detection circuit **25** functions effectively. Therefore, the first lead wire **W1** may also be a lead wire whose resistance value is negligible.

(231) In the first to sixth embodiments, the downstream end of the second voltage resistor **43** included in the second controller **22** does not necessarily have to be connected to the upstream end of the second lead wire resistor **Rw2**, and may also be connected to the ground conductor **G**. In this case, a current output from the second output unit **41** does not flow through the second lead wire resistor **Rw2**, and accordingly, the voltage between the two ends of the second lead wire resistor **Rw2** does not fluctuate in accordance with a current output from the second output unit **41**.

(232) In the first to sixth embodiments, the member connected between the positive electrode of the DC power source **10** and the drain of the second FET **40** is only required to be a resistive member that has a resistance component. Accordingly, the first lead wire **W1** may also be replaced with a circuit element whose resistance value is not negligible. An example of such a circuit element is a semiconductor switch whose ON resistance value is not negligible. Likewise, the member connected between the second drive circuit **42** and the ground conductor **G** in the first to sixth embodiments is only required to be a resistive member that has a resistance component.

Accordingly, the second lead wire **W2** may also be replaced with a circuit element whose resistance value is not negligible.

(233) In the first to sixth embodiments, the switches used to control power supply to the first load **11** and the second load **12** are only required to be semiconductor switches configured in such a manner that the higher a voltage applied to the control terminal is, the smaller a resistance value between the current input terminal and the current output terminal becomes. Accordingly, an IGBT (Insulated Gate Bipolar Transistor) may also be used instead of the first FET **30**. Likewise, an IGBT may also be used instead of the second FET **40**.

(234) The first to sixth embodiments disclosed above are examples in all aspects and should not be construed to be limiting the present disclosure. The scope of the present disclosure is defined not by the above description but by the claims, and is intended to encompass all modifications within the meanings and scope that are equivalent to the claims.

Claims

1. A detection circuit comprising: a current output device configured to output a current corresponding to a circuit voltage between an input terminal of a semiconductor switch to which a current is input and ground of a switching circuit that switches the semiconductor switch ON or OFF; and a circuit resistor through which the current that is output from the current output device flows, wherein a voltage between two ends of the circuit resistor is output, and a voltage between the ground of the switching circuit and a downstream end of the circuit resistor is higher than 0 V; and wherein the current output device includes: a device resistor to which a current flows from the input terminal of the semiconductor switch; and a PNP bipolar transistor including an emitter connected to a downstream end of the device resistor, wherein a potential of a base of the bipolar transistor is a ground potential of the switching circuit, and a current output from a collector of the bipolar transistor flows to the circuit resistor.

2. The detection circuit according to claim 1, further comprising a connection switch connected between the input terminal of the semiconductor switch and the current output device, wherein the current output device outputs a current drawn from the input terminal of the semiconductor switch to the circuit resistor.

3. A power supply control device configured to control power that is supplied via a semiconductor switch, the power supply control device comprising: a switching circuit configured to switch the

semiconductor switch ON or OFF; the detection circuit as set forth in claim 1; and a processor configured to execute processing, wherein the processor instructs the switching circuit to switch the semiconductor switch ON or OFF in accordance with a voltage that is output from the detection circuit, the detection circuit includes: a current output device configured to output a current corresponding to the circuit voltage; and a circuit resistor through which the current that is output from the current output device flows, the detection circuit outputs a voltage between two ends of the circuit resistor, and a voltage between the ground of the switching circuit and a downstream end of the circuit resistor is higher than 0 V.

4. The power supply control device according to claim 3, further comprising a current output unit configured to output a current corresponding to a current flowing through the semiconductor switch, and the voltage between the ground of the switching circuit and the downstream end of the circuit resistor fluctuates in accordance with the current that is output from the current output unit.

5. The power supply control device according to claim 4, wherein the switching circuit steps up the circuit voltage, and applies the stepped-up voltage to a control terminal of the semiconductor switch to switch the semiconductor switch ON.

6. The power supply control device according to claim 4, wherein power is supplied to the switching circuit with a current that flows from the input terminal of the semiconductor switch successively through the switching circuit and the ground of the switching circuit, the processor determines whether or not the circuit voltage is lower than a threshold voltage based on a voltage that is output from the detection circuit while the semiconductor switch is OFF, and if it is determined that the circuit voltage is lower than the threshold voltage, the processor determines whether or not the switching circuit is operating.

7. The power supply control device according to claim 3, wherein the switching circuit steps up the circuit voltage, and applies the stepped-up voltage to a control terminal of the semiconductor switch to switch the semiconductor switch ON.

8. The power supply control device according to claim 7, wherein the processor determines whether or not the circuit voltage is an ON voltage based on a voltage that is output from the detection circuit while the semiconductor switch is OFF, the ON voltage being a voltage that can be stepped up to a voltage with which the semiconductor switch can be switched ON, if it is determined that the circuit voltage is the ON voltage, the processor instructs the switching circuit to switch the semiconductor switch ON, and after giving the instruction to switch the semiconductor switch ON, the processor determines whether or not the semiconductor switch is ON.

9. The power supply control device according to claim 8, wherein power is supplied to the switching circuit with a current that flows from the input terminal of the semiconductor switch successively through the switching circuit and the ground of the switching circuit, the processor determines whether or not the circuit voltage is lower than a threshold voltage based on a voltage that is output from the detection circuit while the semiconductor switch is OFF, and if it is determined that the circuit voltage is lower than the threshold voltage, the processor determines whether or not the switching circuit is operating.

10. The power supply control device according to claim 7, wherein power is supplied to the switching circuit with a current that flows from the input terminal of the semiconductor switch successively through the switching circuit and the ground of the switching circuit, the processor determines whether or not the circuit voltage is lower than a threshold voltage based on a voltage that is output from the detection circuit while the semiconductor switch is OFF, and if it is determined that the circuit voltage is lower than the threshold voltage, the processor determines whether or not the switching circuit is operating.

11. The power supply control device according to claim 3, wherein power is supplied to the switching circuit with a current that flows from the input terminal of the semiconductor switch successively through the switching circuit and the ground of the switching circuit, the processor determines whether or not the circuit voltage is lower than a threshold voltage based on a voltage

that is output from the detection circuit while the semiconductor switch is OFF, and if it is determined that the circuit voltage is lower than the threshold voltage, the processor determines whether or not the switching circuit is operating.

12. A detection circuit comprising: a current output device configured to output a current corresponding to a circuit voltage between an input terminal of a semiconductor switch to which a current is input and ground of a switching circuit that switches the semiconductor switch ON or OFF, wherein the current output device includes a device resistor to which a current flows from the input terminal of the semiconductor switch; and a PNP bipolar transistor including an emitter connected to a downstream end of the device resistor; a circuit resistor through which the current that is output from the current output device flows; and a connection switch connected between the input terminal of the semiconductor switch and the current output device, wherein the current output device outputs a current drawn from the input terminal of the semiconductor switch to the circuit resistor; and wherein a voltage between two ends of the circuit resistor is output, and a voltage between the ground of the switching circuit and a downstream end of the circuit resistor is higher than 0 V; and a potential of a base of the bipolar transistor is a ground potential of the switching circuit, and a current output from a collector of the bipolar transistor flows to the circuit resistor.
